

Queen's University Belfast

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MTH4322: Topological Data Analysis

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1 Summary

Looking at previous examples giving us contributions to the use of TDA and BM, we explored the study done by Swansea University (2020) titled 'Visualizing the Evolution of English COVID-19 Cases with Topological Data Analysis Ball Mapper'. The study aimed to explore the disease spread of COVID-19 among England, looking further than trends and spacial states to higher dimensional interactions between socio-economic data. The study explores the use of Topological Data Analysis and Ball Mapping within this, and provides a strong contribution to their use. Within this investigation, BM was able to highlight where within the character space infection numbers were especially fast rising.

The data was gathered from a variety of sources including the Office of National Statistics (ONS), local authorities and Eurostat. Once data was gathered and summary statistics produced (page 3). The variables in the data, excluding COVID-19 cases, were normalized to an interval of [0,1] to prevent there being one dominant axis.

The methodology for this paper relied on BM to produce and observe multi-dimensional interactions. More specifically, the data was covered with a collection of balls with radius ϵ, with overlapping balls connected by an edge. Effects of noise in the colouration were carefully considered when choosing the epsilon value, attempting to balance this factor with the ability to neatly chart the dynamics of the cases. The colouration of the balls was set to the proportion of England's total cases within each ball. The colour bar was used with lowest values at red and highest at purple, shown in the legend in Figure 1 (page 6). The initial ball map created was then re-coloured on a daily basis, being updated with new case information in Table 2 (page 4). For the later plots (page 7), the outcome was then switched to each of the six variables, used to explore each of the characteristics within the data set for correlations.

Figure 1: Evolution of Cumulative Case Numbers

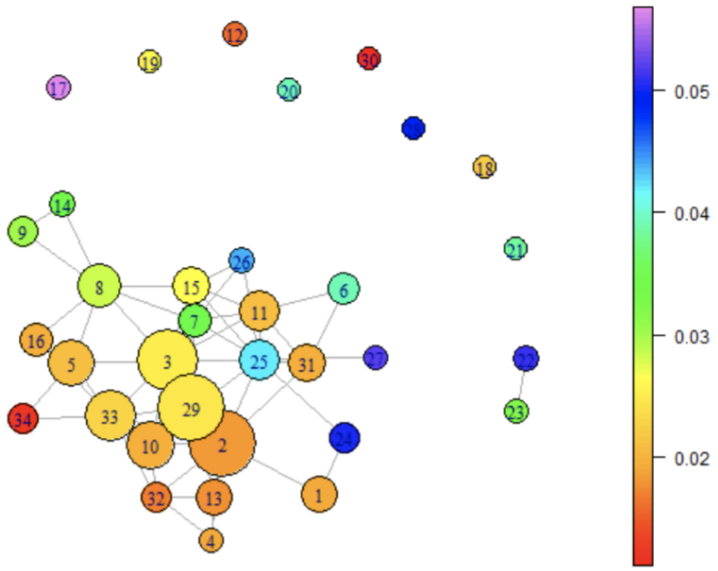


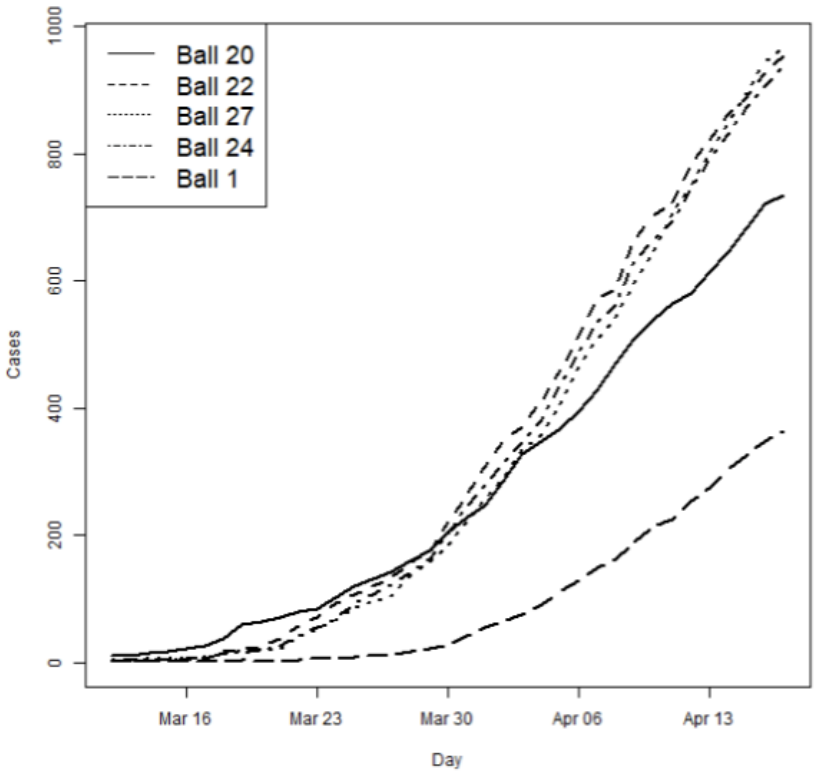
Figure 1: Ball Map of COVID-19 Cases among regions of England

Figure 1 (Labelled figure 1 in the essay) shows the ball map of cumulative cases numbers for the 17th April 2020. The plot shows the difference regions represented in a series of balls, with the colouration showing the proportion of cases, demonstrating the non-uniform spread of cases across clusters. The legend to the right of the plot shows the scaling of the colouration. This along with six other plots were given to represent the changes in the colouration of the data ranging from the 14th March 2020 to 17th April 2020.

The plots showed an initial hot-spot in a high income area within central London (Ball 20), with a dispersion of cases out to lower income areas such as outer London (Ball 22), commuter towns (Ball 27). The spread then made its way further throughout England, eventually reaching northern industrial regions (Balls 9 and 14) . The spread is suggested to have come through other major cities within the country (Ball 7). Although the spread of cases moved to most regions, there was not an entirely uniform spread with commuter cities such as Swindon (Ball 1), remaining to have low cases.

From this the use of BM to show the characteristics within the spread of COVID-19 have been made clear. The maps show the separation of a series of balls with low population density and others with high earnings. More specif-

ically we see ball 20 with purple colouration, which represents areas in central London. Across time the proportion remained consistent, but by April the dominance of ball 20 lessens, with case count increasing overall, leading to an orange colouration across the map. More specifically we see the spread to lower income areas, with some balls even catching up to ball 20 in colouration. Figure 2 (Labelled Figure 3 in Essay) further highlights this, showing a line graph of case numbers over time for specific balls. This graph visualized the increase of case numbers in London boroughs and the eventual flattening of ball 20 with respect to case increase. while this is the case for some balls, a high contrast was seen with ball 1 remaining red while its neighbour ball 24 became more blue. Eventually ball 20 lost its place for the highest cases, replaced by ball 22.



(a) London Boroughs and Commuters

Figure 2: Line graph Showing Evolving Number of Average Cases for Balls of Interest

The change of the colouration to each of the six variables showed the spread of characteristics among the regions. Non-uniformity in case rises was noted, with the right subset showing high increase while the left set saw none, suggesting that confounding factors were at play. The explanation was offered due to a correlation between high median age and low cases. GDP and GVA showed similar spread across the data, with weak similarity with gross pay, suggesting an explanation for factors driving the separation of some balls. Weak correlation between hours worked and COVID-19 cases was found, however this was not explored further.

Looking to the driving forces of these insights, different behaviours in similar balls was highlighted. For example, balls 24 (outer London Districts) and 1 (smaller cities with strong London commuters) showed stark difference in colouration. this is further highlighted by ball 27, containing London boroughs, turning more blue over time. Balls 9 and 14, representing the industrial north, showed low population density yet their colouration became more yellow over time, further highlighting non-uniform changes. these balls were also found to have the highest rate of increase within the data-set. a potential link between major growth balls and these regions was seen through ball 7 (Manchester, Liverpool, Nottingham, Birmingham, Bristol), with an increase in relative proportions through ball 8 to these.
While this research provides useful insights into the spread of COVID-19 within England, the scope of this research is rather small. the data provided only shows insights into England, ignoring other parts of Europe and the UK, and the time-scope for this data ends mid-April 2020, being only a short amount of time into the lockdown. further avenues for this research could be exploring these unaccounted areas, for example looking at the UK as a whole, providing insights into the interactions between the countries.
[1]

[1]To view paper visit: `https://arxiv.org/abs/2004.03282`

2 Our Investigation

Abstract

We apply the Ball Mapper algorithm to European COVID-19 data. The resulting Mapper graphs provide visualisations of the pandemic that, compared to more standard methods, offer a more comprehensive perspective. These graphs reflect the development of the pandemic across Europe and capture both growth rates and the countries most severely affected.

2.1 Introduction

COVID-19 rapidly became established as the global pandemic of our generation. It was and still is causing a large number of deaths, illness and major economic losses in the world but especially in Europe [2]. In the analysis we will use methods of Topological Data Analysis (TDA)[3]. This is a method for understanding data clouds. [4] that gain insight into the data by treating it as a geometric object. Information can then be extracted based on its shape.
For this report we will build on the recent development presented in Dlotko (2019a) to apply the Ball Mapper algorithm to epidemiology[5] specifically to map the spread of COVID-19 across countries in Europe, monitoring how they reacted to the second wave and analysing if the socio-economic factors contributed to COVID-19 spread.

2.2 Data

For the report we summarized, we chose to standardize all non-COVID variables. The following table presents a summary of these variables.

Characteristic	**Median**	**Std. Dev.**	**Min**	**Max**
Population density (per km^2)	102	2885.46	3.9	19000
GDP per Capita ($)	24221	40032.71	3720	231600
Gross Value Added (million USD)	106553	849844.5	5253	3886060
Average Hours Worked (per year)	1700	239.90	1372.04	2252
Annual Gross Income (€)	38600	24615.57	8400	90000

Table 1: Updated summary statistics for key economic and demographic indicators.

Note: This data is of European countries although the following have been omitted as they have caused the data to skew and lack of concrete statistics.

- Vatican City
- Andorra
- San Marino
- Liechtenstein

As one can observe there is major variation between the ranges of variables as they are using varying systems of measurement. This means these variables will take on disproportionate weights in determining the shape of the Mapper graph. We therefore need to offset this by normalising our data.

2.2.1 Normalisation

To ensure comparability across variables measured on different scales, each variable was normalized to the unit interval $[0, 1]$. For a given variable x, the normalized value $\hat{x}$ is defined as:

$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

where $\min(x)$ and $\max(x)$ denote the minimum and maximum values of x across all observations. This transformation rescales the data so that $\hat{x} \in [0, 1]$, preventing disproportionate weighting in the Mapper graph.

[2] For more information, see `https://www.sciencedirect.com/science/article/pii/S0264999323003632`

[3] Started from the works of Carlsson (2009) and Nicolau et al. (2011)

[4] A data cloud refers to a collection of data points in high-dimensional space, which can be studied using geometric and topological methods.

[5] Epidemiology studies patterns, causes, and effects of health conditions in populations, guiding prevention, control, and public health decisions.

2.2.2 Log transformation

From the summary table above we can see that for the variables, Population density, GDP per capita and Gross value added and Average annual gross income. The standard deviation is considerably high with the different in minimum and maximum values meaning our data will be skewed. We can take population density as an example where Malta is the second most dense country in Europe with 1686 compared to Monaco's 19000 (see section 4.2)
Therefore a log transformation must be used to reduce skewness and compress outliers. This helps the variables approximate normality and prevents dominant regions from distorting the typology. So for each variable X, the log transformation is given by

$$X_i^* = \log(X_i + 1), \quad i = 1, 2, \ldots, n$$

2.3 The Ball Mapper Algorithm

The Ball Mapper algorithm is a tool used to produce a representation of a high dimensional dataset, visualising and exploring it in two dimensions.
To achieve this aim, an input collection of points P together with a distance function d, is covered with a collection of balls of radius $\varepsilon > 0$. The fact that the whole point cloud P is covered by balls of radius ε centred at

$$c_1, c_2, \ldots, c_n$$

implies that for every $x \in P$ there exists some $i \in \{1, \ldots, n\}$ such that

$$d(x, c_i) \leq \varepsilon.$$

Given such a cover, an abstract graph is constructed in the following way:

- Vertices correspond to the centres of balls $c_1, \ldots, c_n$.
- An edge is placed between vertices c_i and c_j for $i, j \in \{1, \ldots, n\}$ if there exists a point $p \in P$ such that

$$d(p, c_i) \leq \varepsilon \quad \text{and} \quad d(p, c_j) \leq \varepsilon.$$

In other words, the point p is jointly covered by the ball centred at c_i and the ball centred at c_j.

3 Results

The following table 3 presents the country assignments for each Ball ID, establishing the link between the abstract topology of the Ball Mapper graphs and the concrete geographic units they represent.

Ball ID	Countries
1	Albania
2	Austria, Belgium, Denmark, Finland, Portugal, Spain
3	Belarus, Serbia, Ukraine
4	Bosnia and Herzegovina, Montenegro, North Macedonia, Serbia
5	Bulgaria, Serbia
6	Croatia, Greece, Hungary, Romania
7	Cyprus, Estonia, Slovakia
8	Czechia, Greece, Hungary, Poland, Portugal, Romania, Slovakia
9	France, Germany, Italy, Spain, United Kingdom
10	Iceland, Norway
11	Ireland
12	Kosovo
13	Estonia, Latvia, Lithuania, Slovakia, Slovenia
14	Luxembourg
15	Malta
16	Moldova, North Macedonia
17	Monaco
18	Belgium, Denmark, Netherlands, Switzerland, United Kingdom
19	Russia
20	Finland, Lithuania, Norway, Slovenia, Sweden
21	Turkey

Table 2: Ball to Country Mapping (applies identically across all waves)

3.1 Ball Mappers

3.1.1 Start of Wave 1

Figure 3 shows a ball map of the initial wave of COVID-19 starting the 20th March 2020. It visualises the number of COVID-19 cases within different regions of Europe. The clusters here represent countries that have very similar socioeconomic and demographic profiles at the time i.e, median age, GDP per Capita, population density average gross pay and hours worked etc.

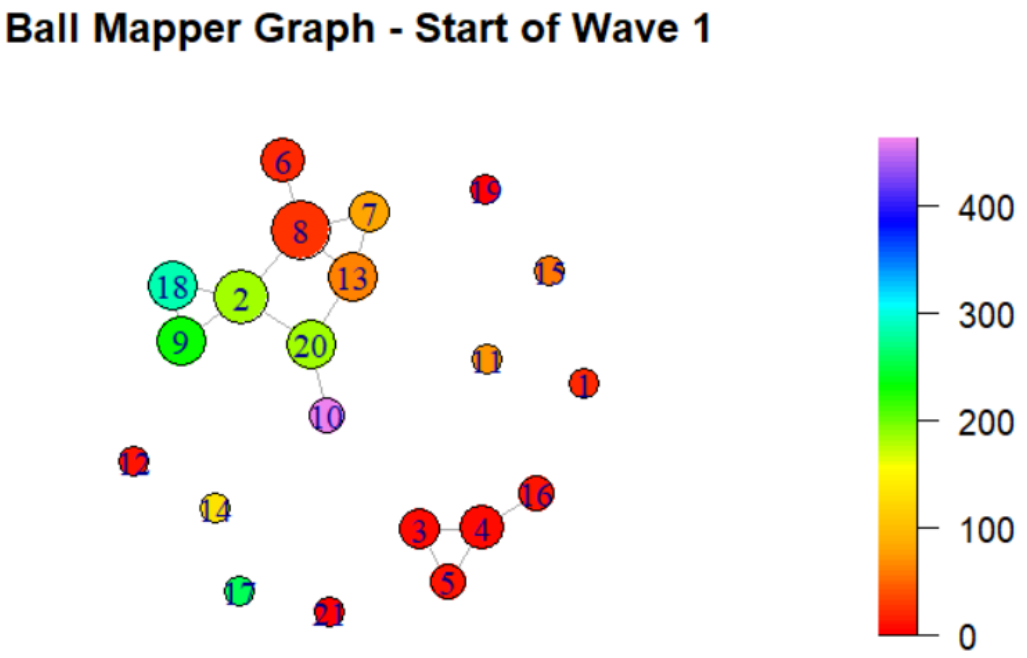


Figure 3: Ball Mapper Graph – Start of Wave 1 (20th March 2020). This graph visualizes the initial clustering of COVID-19 cases across regions.

Initial Analysis We see multiple clusters within the graph, one with a difference in colouration. There are also multiple outliers of note, including some with extremely high COVID-19 case numbers.

From the plot we see a large cluster in the top-left. This cluster shows a variety of colouration within itself. The notably high regions within this cluster are Scandinavian regions of Europe, containing countries such as Iceland, Norway, Denmark and Switzerland. This is of note as these countries reacted promptly and harshly to the initial outbreak of COVID-19, however the case numbers remain high. These countries have high GDP per capita in comparison to the other countries with the cluster. Looking at the edges connecting the balls within this cluster, we find potential avenues to keep an eye on as the data evolves. There is a connection between balls (8) and (19), and this is through both containing the UK. We also see the connection between the northern European region and the west. This is shown through balls (2) and (13), with (2) connecting via Spain and Belgium to Finland and ball (13) connecting via Lithuania to Slovakia. This cluster is of note for further investigation as it shows the difference in behaviour within regions of similar characteristics, which may suggest confounding factors at play.

Ball (17) also stands out as having high numbers of COVID-19 cases. This ball represents Monaco. This is of note as Monaco reacted pre-emptively to COVID-19, reducing public space usage and implementing measures against the spread of COVID-19 earlier than its neighbours. This ball remains a notable outlier due to its extremely high GDP in comparison to other countries within Europe. other outliers found, such as balls (1) and (11), representing Albania and Ireland, may be a result of low population density or other factors that don't align with neighbours.

Another cluster can be seen in the bottom right of the plot, showing balls with notably low COVID-19 cases indicated by its red colouration. The balls within this cluster represent countries with low GDP per capita in middle southern Europe, such as Montenegro, Bulgaria and Bosnia. These are called the Western Balkan countries. The connected edges within this clump come from a shared countries amongst each ball, namely Serbia. This is of note as Italy, a neighbouring country to this region, faced the highest initial outbreak of COVID-19 cases within Europe.

3.1.2 Beginning of Wave 2

Figure 4 shows the cumulative COVID-19 cases from the 24th August 2020. This approximately marks the end of the first wave of COVID-19 and the beginning of the second COVID-19 wave.

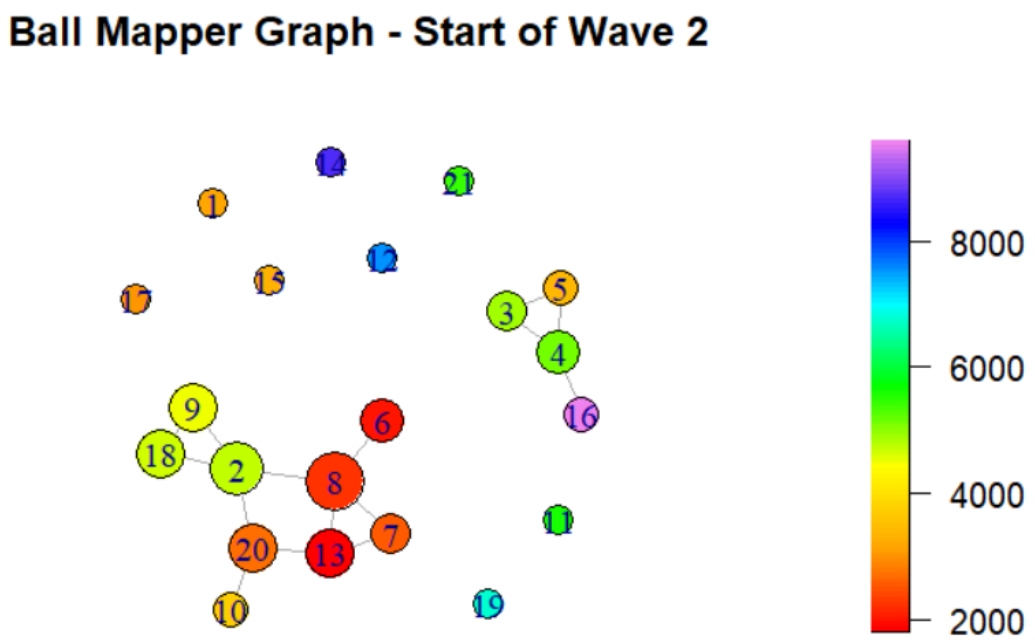


Figure 4: Ball Mapper Graph – Start of Wave 2 (24/08/20). This graph shows the early structure of the second wave of COVID-19 spread.

Initial Analysis From this graph immediately we can make some interesting inferences on how these countries had been affected by COVID-19 at the beginning of the second wave:

Figure 4 shows how the pandemic developed as it continued its course. Of particular note, ball 16 is at the top of our colour scale with the highest cumulative cases at this stage. This ball represents Moldova and North Macedonia, which share very similar socioeconomic profiles. Due to their poorer economies, they were unable to adequately respond to the rising case numbers. We also observe that North Macedonia is found in ball (4) as well, which indicates that Moldova was the biggest contributor to the poor case rate. This interpretation is further supported by UN Moldova, which recorded a "record high for cases in mid-August 2020".[6] This differs from ball four which includes some mid income western Balkan nations such as Montenegro and North Macedonia. We see that they in comparison have much fewer cases of COVID-19 despite also having similarly low income economies. The best performing out of this cluster is ball (5) which includes Bulgaria which is a much wealthier economy compared to the others in the region. They at this stage had the lowest cases in that region.

If we then look at the main largest cluster of balls i.e, (9,18,2,...) we see that it has much more moderate COVID cases at this stage of the pandemic with the economies of these countries being more affluent and having better testing and research for their own populations. Ball (2), Austria, Belgium etc. have around 5000 cases per million at this stage they are not doing as well when compared to larger balls such as 8 which includes Greece, Hungary etc.

Overall Analysis To compare the change in figure4 to figure3 we will look at changes in the colouring and how the countries reacted to the COVID pandemic's first wave in Europe. The Western Balkan countries (cluster 3,4,5,16) had a very large change in colouring as we see that they go from very few cases in figure3 to having the largest number in the case of ball (16). These countries did not respond well to the increase in cases.

The larger cluster (9,18,12...) we see how they, despite having among the largest case numbers at the beginning of the pandemic maintain a reasonably low level of cases. Such as ball 10, Iceland and Norway who very impressively bring their cases from being the highest in Europe to comparably lower end of cases by the time the end of the first wave has come. We see a similar picture in ball 18 where countries such as the United Kingdom are. They initially had higher cases than most to begin with but by the time we reach this stage the 24th of August they have manages to stabilise cases to a level comparably low compared to those Balkan nations, (16,4,3,5).

[6]For more details see `https://share.google/lO9J0fbyZ4Ee8Fyb0`

3.1.3 Ending of Wave 2

Figure 5 shows the cumulative COVID-19 cases from the 21st of April 2021. This roughly marks the end of the second COVID-19 wave as countries where easing lockdown measures and death and case tallies were dropping.

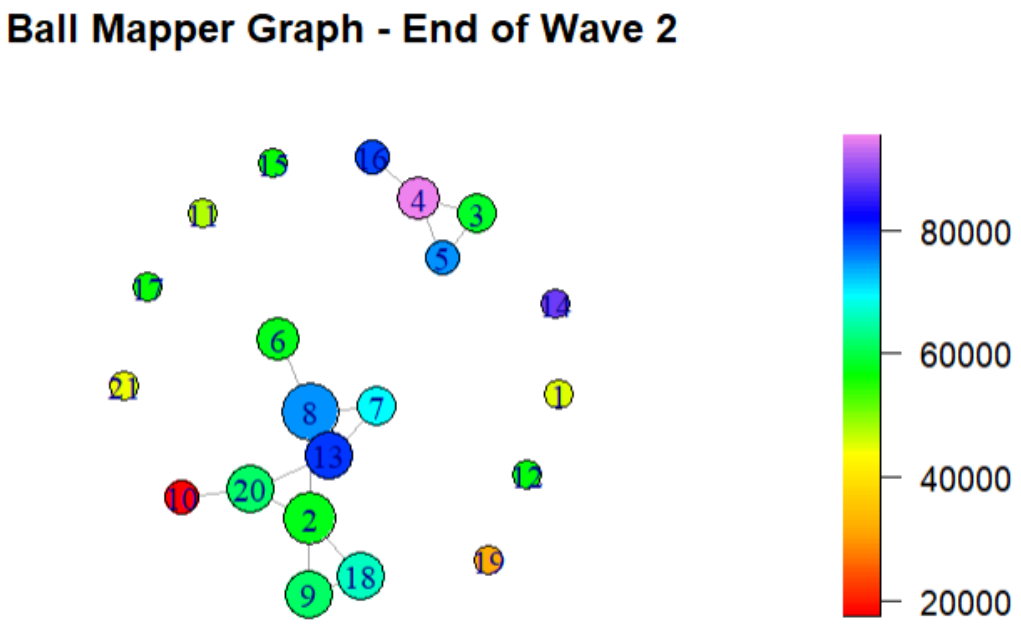


Figure 5: Ball Mapper Graph – End of Wave 2. This graph captures the clustering pattern at the end of the second wave.

Initial Analysis From this graph immediately we can make some interesting inferences on how these countries had been affected by COVID-19 at the end of the second wave:

Iceland and Norway (ball 10) have an incredibly low cumulative case rate compared to other balls they are linked with, most notably ball 20 which includes Norway. We can say that Iceland was the most effective at stopping the spread of COVID-19 out of the European countries we studied. This is due to multiple factors, the most obvious being that it is a remote island and therefore can monitor all those who enter and exit the country thoroughly. In addition, the Icelandic authorities authorised an app to help track infections of COVID-19 more effectively.[7] In our 4 ball cluster (balls 16, 4, 3 and 5) We see the majority of these are Western Balkan countries. These countries all performed poorly compared to the majority of other European countries. This is most notably due to crash in the economy for these countries due to the COVID-19 outbreak. In the World Bank Groups study of this they said the following "In the downside scenario, growth in the Western Balkans would plummet by about 5.7 percent in 2020, causing a more severe recession than the global financial crisis." [8]

Ball 19 (Russia) has a relatively low cumulative COVID-19 case rate compared to other countries that are also unique in terms of socio-economic status. Although this could be due to the lack of testing that Russia has performed. On April 12th 2021, Russia's testing rate per 1000 people was over 4 times less than countries like Denmark.

Overall Analysis Comparing to the start of wave 1 we can see how effective Iceland's strategy was in mitigating the spread of COVID-19, even though they would've found it easier to track people coming in and out of the country. We can see that balls 8 (Czechia, Greece, Hungary, Poland, Portugal, Romania, Slovakia) and 13 (Estonia, Latvia, Lithuania, Slovakia, Slovenia) ,or the lesser economically developed central European countries, struggled when compared to early wave 1 COVID-19 cases. This is due to a low vaccine coverage at this time. For example by September 2021 only 37 % of Latvia's population was fully vaccinated.[9]

3.2 Limitations

The research undertaken aimed to explore the COVID-19 rates within Europe, however when carrying out the data collection process, there were countries within the data-set that proved to be significant outliers, skewing the data set. This resulted in the removal of these countries. while this avoided skewness of the data, it was resulted in a incomplete analysis of the aim of this research, as not all countries within Europe have been mapped.

[7] For more information see `https://guidetoiceland.is/travel-info/how-iceland-became-safe-from-the-covid-19-coronavirus`. Within days, 40% of the population had downloaded this app, showcasing the country's collective effort.

[8] For more reading, see`https://share.google/0YJRXrna4Ar1JdhXJ`

[9] For more reading see `https://stat.gov.lv/en/statistics-themes/social/health-status-and-morbidity/press-releases/8854-vaccination-against-covid` and`https://bnn-news.com/lithuania-and-estonia-report-more-than-half-of-their-population-are-vaccin`

Another limitation to this research is that the finding gathered are under the assumption that all countries within Europe test and record the COVID-19 rates to the same extent. For example, some countries may be more relaxed when recording COVID-19 cases, resulting in a misleading statistic produced for that country. this would affect the accuracy of our findings as it would be based on biased data, having weight towards countries that record COVID-19 cases more strenuously.

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